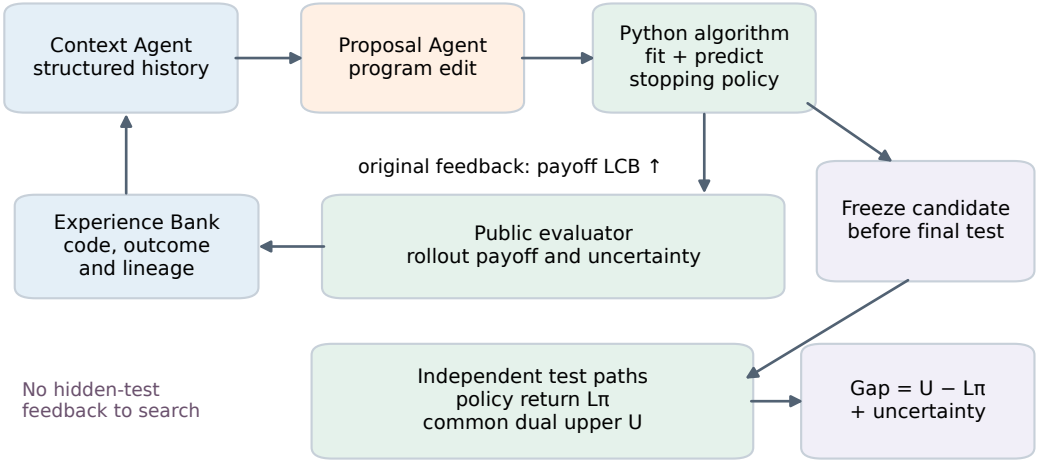
Final holdout: primal–dual results

Method / selection rule	L / K	U / K	(U − L) / K	Gap score	SD (seeds)
Ridge	0.151048	0.255374	0.104326	0.185268	0.096560
PCA + Ridge	0.151240	0.255374	0.104134	0.185036	0.096206
Gated Ridge	0.150340	0.255374	0.105034	0.186014	0.095974
MLP	0.141704	0.255374	0.113670	0.195599	0.092917
Ridge + MLP	0.150750	0.255374	0.104624	0.185697	0.095675
Threshold policy	0.058579	0.255374	0.196795	0.289598	0.073307
Policy iteration	0.151240	0.255374	0.104134	0.185036	0.096206
Original Context: best generated	0.149877	0.255374	0.105497	0.186558	0.095784
Original direct: best generated	0.150137	0.255374	0.105237	0.186371	0.095491
Original Context incumbent	0.150543	0.255374	0.104830	0.185904	0.095173
Original direct incumbent	0.150415	0.255374	0.104959	0.186041	0.096057
Structured-memory incumbent	0.151048	0.255374	0.104326	0.185268	0.096560
Score-only incumbent	0.150679	0.255374	0.104695	0.185653	0.095898

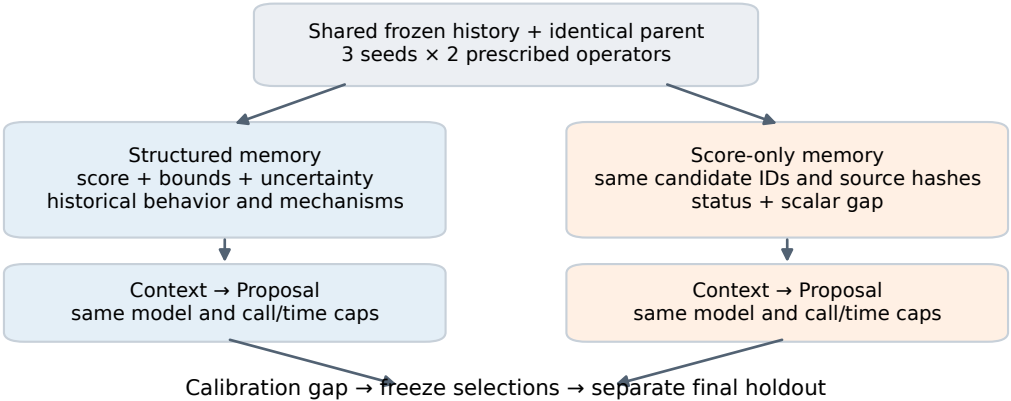
Three seed blocks; four test contracts per block. Means of normalized sample estimates.

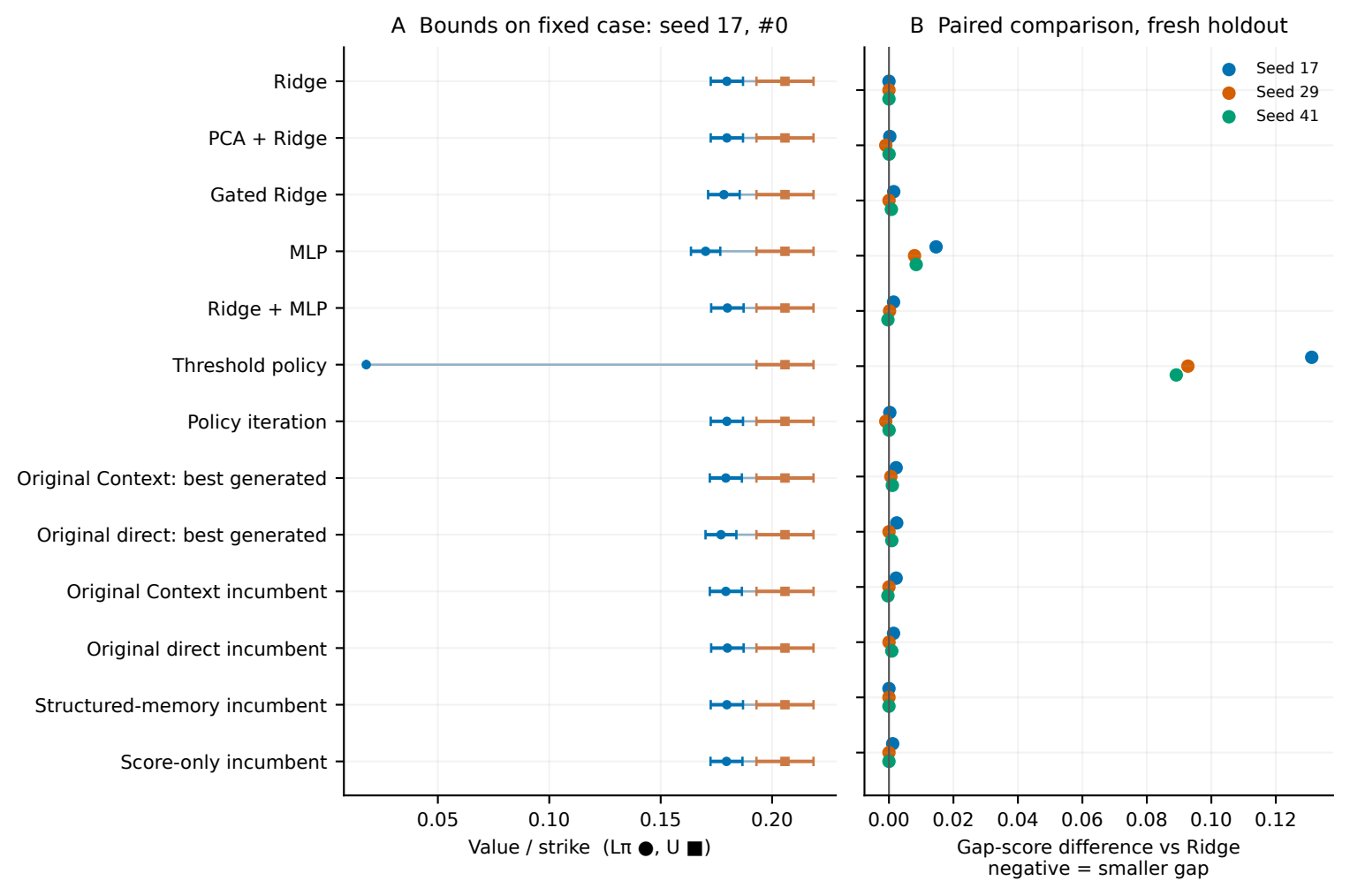
Gap score includes Monte Carlo uncertainty and a tail term. SD is across seeds; selections froze before testing.

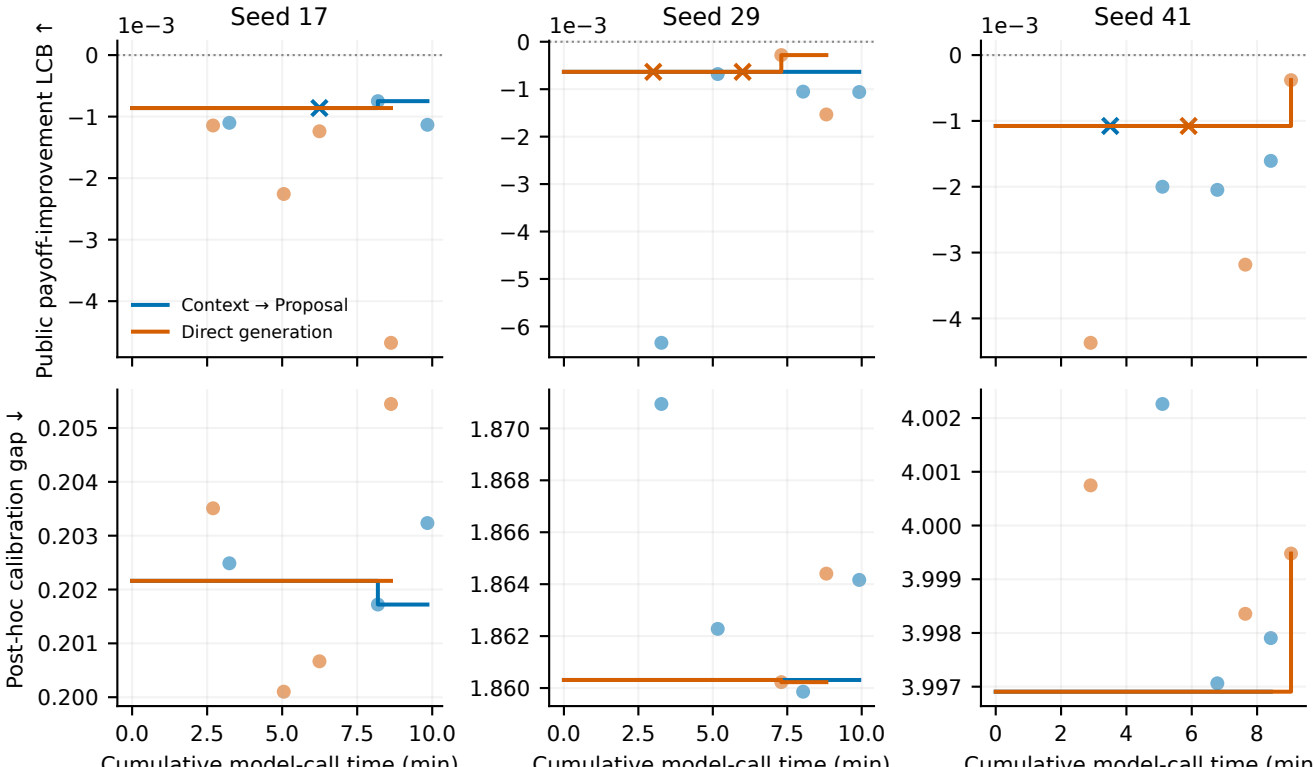
A Search loop and independent gap evaluation

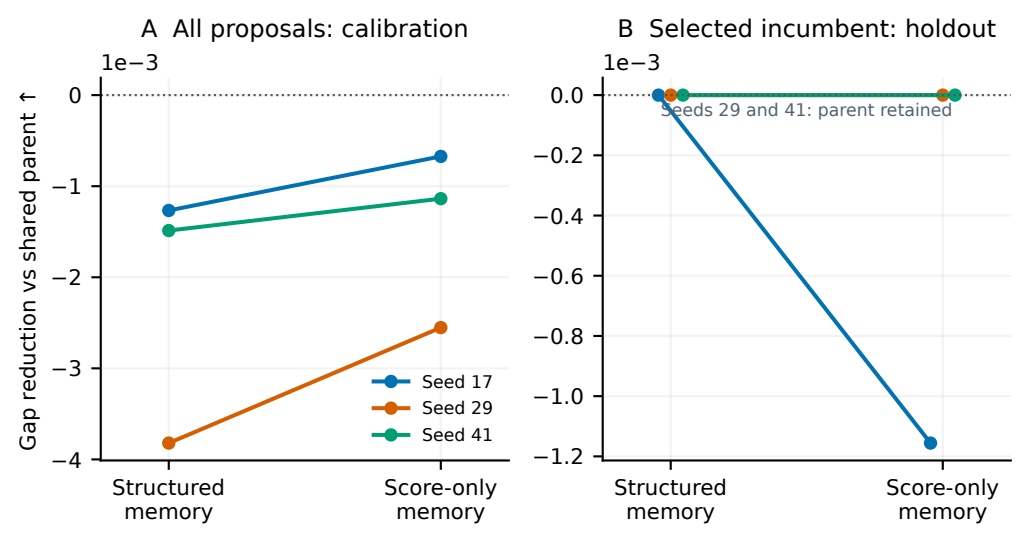


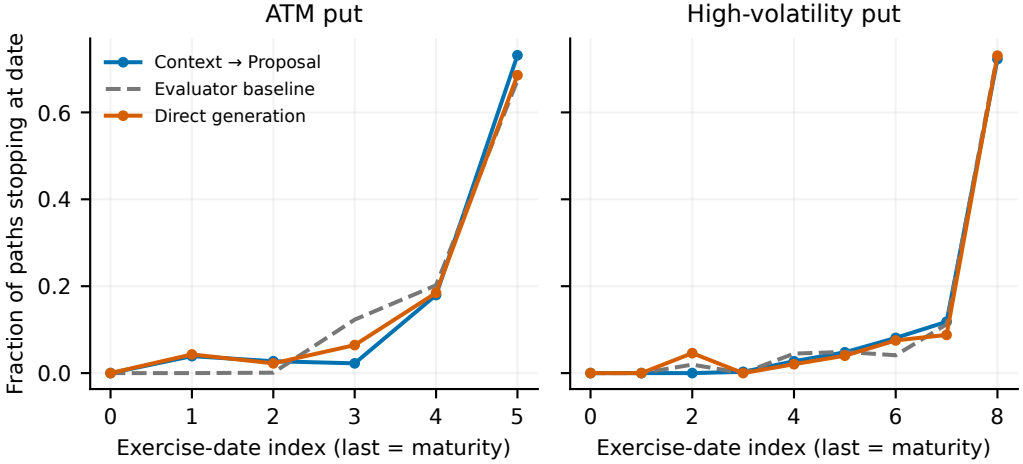
B Matched feedback ablation

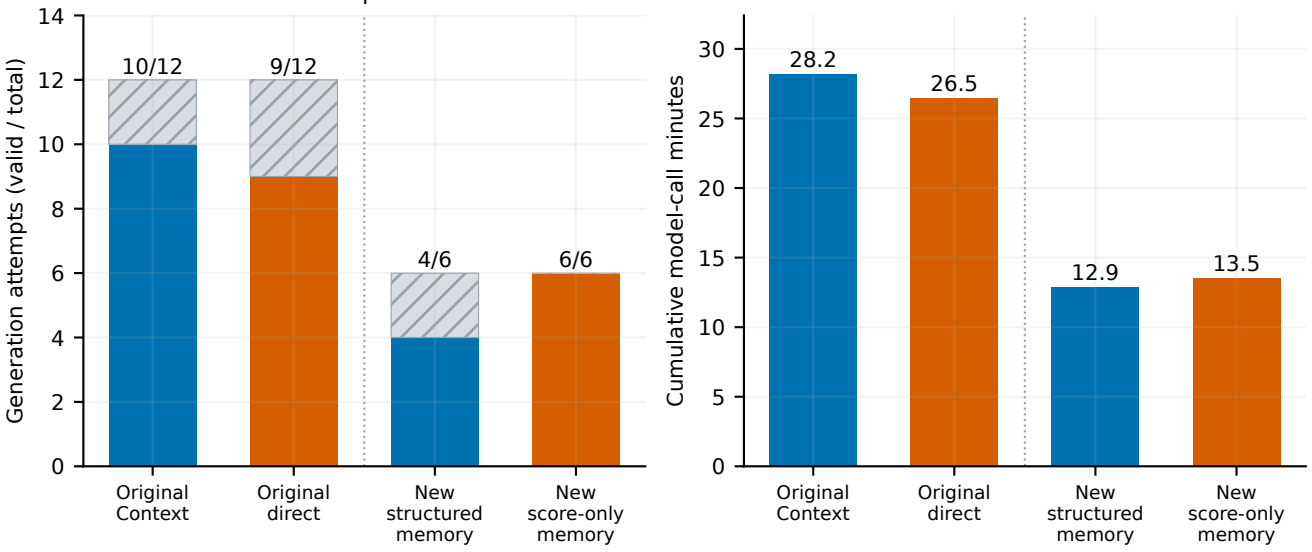



A Bounds on fixed case: seed 17, #0
B Paired comparison, fresh holdout
Seed 17
Seed 29
Seed 41
Ridge
PCA + Ridge
Gated Ridge
MLP
Ridge + MLP
Threshold policy
Policy iteration
Original Context: best generated
Original direct: best generated
Original Context incumbent
Original direct incumbent
Structured-memory incumbent
Score-only incumbent
0.05
0.10
0.15
0.20
Value / strike (Lπ ●, U ■)
0.00
0.02
0.04
0.06
0.08
0.10
0.12
Gap-score difference vs Ridge
negative = smaller gap


Original search: candidates and seed-inclusive incumbents
Seed 17
Seed 29
Seed 41
1e−3
Public payoff-improvement LCB ↑
Context → Proposal
Direct generation
Post-hoc calibration gap ↓
Cumulative model-call time (min)


A All proposals: calibration
1e−3
0
−1
−2
−3
−4
Gap reduction vs shared parent ↑
Seed 17
Seed 29
Seed 41
Structured memory
Score-only memory
B Selected incumbent: holdout
1e−3
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.2
Seeds 29 and 41: parent retained
Structured memory
Score-only memory


ATM put
High-volatility put
Context → Proposal
Evaluator baseline
Direct generation
Fraction of paths stopping at date
0.0
0.2
0.4
0.6
0
1
2
3
4
5
6
8
Exercise-date index (last = maturity)
Exercise-date index (last = maturity)


A All attempts retained
Generation attempts (valid / total)
14
12
10
8
6
4
2
0
10/12
9/12
4/6
6/6
Original Context
Original direct
New structured memory
New score-only memory
B Recorded model cost
Cumulative model-call minutes
30
25
20
15
10
5
0
28.2
26.5
12.9
13.5
Original Context
Original direct
New structured memory
New score-only memory